IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Received 22 July 2025, accepted 3 September 2025, date of publication 10 September 2025, date of current version 17 September 2025.

*Digital Object Identifier 10.1109/ACCESS.2025.3608241*

RESEARCH ARTICLE

# Aerospace Bricks: Modular and Scalable 2D Cellular Automaton Framework for Delay and Conflict Analysis in En-Route Airspace

**KATSUHIRO SEKINE[1], TOMOAKI TATSUKAWA[2], (Member, IEEE), YOSHIAKI KAWAGOE[3,4], AND ERI ITOH[1,3]**

[1]Research Center for Advanced Science and Technology, The University of Tokyo, Tokyo 153-8904, Japan
[2]Department of Information and Computer Technology, Tokyo University of Science, Tokyo 162-8601, Japan
[3]Department of Aeronautics and Astronautics, The University of Tokyo, Tokyo 113-8654, Japan
[4]Department of Aerospace Engineering, Tohoku University, Sendai, Miyagi 980-8579, Japan

Corresponding author: Katsuhiro Sekine (k-sekine@g.ecc.u-tokyo.ac.jp)

This work was supported in part by Japan Society for the Promotion of Science (JSPS) KAKENHI under Grant 24KK0088 and Grant 25K23471, and in part by Ministry of Education, Culture, Sports, Science and Technology (MEXT) as "Program for Promoting Researches on the Supercomputer Fugaku" and used the Computational Resources of the Fugaku Supercomputer provided by the Institute of Physical and Chemical Research (RIKEN) Center for Computational Science (Project ID: hp250221) under Grant JPMXP1020230215.

**ABSTRACT** Operational delays and air traffic complexity have increased with the growing global demand for air traffic, leading to irregular and less predictable traffic patterns. Although existing air traffic models have effectively addressed these challenges, they often trade off scalability, realism, and generalizability, particularly in en-route airspace, where both horizontal and vertical maneuvers matter. This paper proposes a two-dimensional cellular automaton model for simulating aircraft behavior in both the route direction and flight level. The model integrates realistic phase-specific aircraft dynamics and selectively applies separation controls (e.g., vectoring, holding, and speed control) only when operationally relevant. The proposed model is part of the Aerospace Bricks framework, which uses minimum simulation units ("bricks") to represent specific airport and airspace features. Two case studies were conducted. The first used 10-day arrival traffic data from eight major Japanese airports to validate the model's fidelity. The model successfully reproduced typical vector patterns and agreed closely with actual flight distances with spacing delays, averaging 95.3 s per flight within 80 NM of Tokyo International Airport. The second study simulated a full day of traffic in the Fukuoka Flight Information Region and identified high-complexity upper sectors at FL335 and above. The analysis detected several hotspots, such as flights with spacing adjustments and flights with traffic conflicts. These results demonstrate that the proposed model can efficiently reproduce realistic traffic dynamics and quantify delay and conflict risks. The proposed framework enables users to configure simulation environments that are easily tailored to specific operational scenarios.

**INDEX TERMS** Airspace complexity, air traffic control, air traffic flow, air traffic management, air traffic modeling, cellular automata, cellular automaton, conflict resolution, en-route simulation, lattice-based dynamics.

## I. INTRODUCTION

Air transport serves as a critical enabler of economic growth and social development by supporting global connectivity, international trade, and tourism. After experiencing a sharp decline due to the COVID-19 pandemic, the aviation industry has shown a strong recovery, with global passenger numbers reaching 4.5 billion and total industry revenues reaching US $908 billion in 2023 [1]. Passenger traffic is projected to double by 2040 [2], which highlights the sector's long-term growth trajectory. Despite this resurgence, however, the aviation sector faces persistent operational constraints. One

The associate editor coordinating the review of this manuscript and approving it for publication was Yanli Xu.

© 2025 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License.
For more information, see https://creativecommons.org/licenses/by/4.0/

of the most pressing challenges is the continued dependence on legacy infrastructure and outdated operational procedures, which significantly contribute to systemic inefficiencies, such as flight delays. For example, in 2019, 19.9% of the flights in the United States and 23.5% in Europe were delayed by more than 15 min [3].

In addition to congestion and delays, increasing attention is being paid to the growing complexity of air traffic, which poses a significant challenge to air traffic management (ATM) [4]. This complexity is particularly evident in the context of recent operational concepts such as free-route airspace (FRA) [5], which allows aircraft to select more flexible user-preferred trajectories. Although this approach improves route efficiency and flight autonomy, it also leads to more heterogeneous traffic flows, characterized by greater variability in aircraft headings, speeds, and altitudes [6]. These irregular and less predictable patterns can intensify the workload of air traffic controllers (ATCos), not necessarily because of the number of aircraft, but because of the increased frequency of interactions and potential conflicts between trajectories [7]. For example, crossing a single aircraft through a cluster of flights already cruising in the same direction with safe separation can substantially increase the cognitive demands and operational workload of ATCos.

To address these challenges, previous studies have explored a range of methods for modeling and analyzing air traffic dynamics, from continuous fluid-like representations to detailed simulations of individual aircraft behavior. Compared to data-driven models such as machine learning [8], these physics-based models offer explicit representations of traffic dynamics that remain valid even outside the scope of past data. This mechanistic foundation allows them to capture emergent behaviors and evaluate hypothetical scenarios that data-driven methods alone may not adequately address. Therefore, these models have contributed significantly to the understanding of congestion patterns, controller workloads, and system-wide performance in various traffic scenarios. However, many of these models either abstract aircraft-level behavior to retain tractability or incorporate detailed agent-based decision logic at the cost of scalability and generalizability. Although cellular automaton (CA) models offer a promising middle ground by combining distinct behavioral rules with scalable grid-based representations, most existing CA applications remain domain-specific. They often rely on simplified assumptions such as one-dimensional (1D) routing structures, fixed flow directions, or region-specific procedures. These limit their ability to capture complex interactions in en-route airspaces with flexible routing and mixed traffic. Developing models capable of realistically simulating delays and conflicts with high computational efficiency can facilitate more accurate performance assessments and contribute to the design of adaptive flow management strategies.

Therefore, the authors developed a nationwide air traffic simulation by using a one-dimensional cellular automaton (1D-CA) model [9] that captures aircraft flow dynamics with only a small set of rules and parameters. Despite its structural simplicity, the model successfully reproduces arrival delays and congestion patterns at major airports, such as Tokyo International Airport (RJTT) [10], and has been applied to the design and evaluation of arrival management systems in Japanese airspace [11], [12]. Building on this foundation, we introduce the concept of *Aerospace Bricks*, a modular simulation framework that allows air traffic systems to be constructed from reusable, fine-grained components. In this framework, each ''brick'' represents a minimal, self-contained simulation unit (e.g., traffic flow in a sector, taxiway operations), which can be combined flexibly to match the desired spatial scale and analytical scope. This design not only overcomes the domain-specific limitations of prior CA studies but also enables rapid prototyping and cross-regional benchmarking of emerging ATM concepts under consistent methodological assumptions. Furthermore, it serves as a flexible experimentation platform for integrating advanced approaches, such as machine learning, digital-twin technologies, and hybrid human-in-the-loop scenarios.

As the first step in this project, a novel two-dimensional cellular automaton (2D-CA) model is proposed that explicitly represents both horizontal and vertical aircraft movements within en-route airspace. The model is designed to quantify the delays resulting from traffic bottlenecks while also assessing the potential conflicts that may occur at road intersections or merging points. By incorporating phase-specific aircraft behavior (e.g., climb, cruise, and descent) and defining separation-based interaction rules, the model captures the essential dynamics of conflict emergence without imposing computationally expensive full-resolution control logic. The core contribution of this study is the development of the foundational engine of the framework and defining standardized interfaces that enable users to easily configure simulation environments tailored to specific operational scenarios.

The remainder of this paper is organized as follows. After reviewing the literature in Section II, we provide an overview of the 2D-CA-based air traffic model in Section III. Section IV presents Case Study I, which validated the proposed model using arrival traffic data from eight major airports in Japan. Section V presents Case Study II, a global analysis of air traffic patterns in the Fukuoka Flight Information Region (FIR). Finally, Section VI discusses the key findings and their implications for ATM practices, and Section VII concludes the paper.

## II. CURRENT STATE OF THE ART

### A. MODELING AIR TRAFFIC FLOW

Air traffic flow has been modeled using two major approaches: Eulerian (macro-level) models, which treat aircraft as continuous flows, and Lagrangian (meso/micro-level) models, which track individual aircraft. Macro-level models based on equations (e.g., partial differential equations [13] and discrete-time difference equations [14], [15], [16])

describe the airspace using flow and density conservation equations over spatial cells, whereas micro-level models (e.g., agent-based models [17], [18], [19]) capture the behavior and decision-making of individual aircraft or agents. Although macro-level models are computationally efficient because they depend on spatial discretization rather than the number of aircraft, a key limitation is capturing aircraft-level behaviors such as merging/dividing or stochastic weather effects [20]. The agent-based model (ABM) and multi-agent system (MAS) can naturally capture the autonomy of pilots, airlines, and controllers as agents, enabling the analysis of phenomena emerging from their interactions [21], [22]. However, ABM/MAS requires many assumptions (e.g., agent decision rules and communication protocols) and extensive calibration, which can lead to high computational costs, particularly for large networks. Moreover, granting a high degree of autonomy to agents often leads to a combinatorial increase in the number of parameters, which in turn tends to limit the size of the airspace that can be realistically simulated.

Microscopic models, such as queuing networks and cellular automata, simulate aircraft-level interactions based on simplified rules or flow assumptions without relying on fully autonomous agent decision-making. Queuing models represent congestion phenomena such as departure queues on airport system [23], airport runways [24], radar vectors in the air [25], [26], and conflicts in airspace [27]. Although queuing models are scalable and relatively easy to implement, their accuracy depends heavily on model assumptions, such as the distribution type and process structure. Consequently, they may fail to capture extreme or atypical operational scenarios, particularly under highly uncertain or nonstationary conditions. CA provides a distinct rule-based framework for simulating air traffic as agents interacting in a gridded space. Each cell represents a small volume or waypoint, time advances in fixed steps, and the aircraft moves between cells according to local transition rules. Compared to more complex models such as Eulerian models, CA models are computationally friendly and allow real-time simulations that are easy to interpret [20]. Although continuous flight paths and 4D trajectories are approximated by discrete cells and precise ATC procedures may be abstracted away, these CA models have shown reasonable realism within simplified settings (e.g., capturing macroscopic flows and queuing patterns).

### B. CELLULAR AUTOMATON MODEL

The application of CA models to air-traffic flow was pioneered as a means of exploring novel strategies to improve system performance and airspace capacity [28]. Subsequent work extended this concept by developing a 2D-CA-based framework that employed a Eulerian approach to simulate sections of the European en-route airspace and investigate capacity constraints [29]. That model defines each cell as an airspace sector with a predefined capacity constraint. Aircraft are allowed to enter a sector only when the current occupancy is below this limit. Upon reaching capacity, the sector denies entry, resulting in a delay of one time step for the aircraft. After experiencing two successive delays, the aircraft is eligible for rerouting to an adjacent sector where capacity remains available.

A 2D-CA model employing a Lagrangian framework was proposed in [30] to represent surface operations at the Tokyo International Airport (RJTT), in which long-range interactions among aircraft were considered using a variable taxi speed based on a 2D Nagel–Schreckenberg (NaSch) model. In [31], a 1D-CA model was developed to realistically replicate the dynamics of aircraft movements on the ground. That model was used to evaluate modifications to airport layouts, including the addition of taxiways and the optimization of taxi routes. In [32], the surface traffic at Fukuoka International Airport (RJFF) was modeled using 2D-CA with a volume exclusion rule to avoid conflicts. By updating this CA, 2D-CA models surface traffic management, which includes arriving, taxiing, parking, and departing aircraft, and mutual interactions [33].

For the Lagrangian framework for in-flight traffic, a 2D-CA model was proposed in [34] to represent an en-route self-spacing autonomous flight system. In that model, each cell was limited to hosting at most one aircraft per timestep. The framework assumes that each aircraft knows the current and predicted future positions of all other aircraft in the airspace. For real-time aircraft landing schedules in terminal maneuvering areas, 1D [35] and 2D [36] CA models have been proposed. In [37] and [38], a 2D-CA model was developed to find flight paths that avoid weather-restricted airspaces along with optimization techniques. Similarly, CA modeled the standard terminal arrival routes (STAR) around Atlanta International Airport on 1D [39] and 2D grids [40] (divided by the altitude and direction of the flight path), where a safety distance parameter governs whether a cell ahead is free for an aircraft. A similar approach was used in [41]; however, it employed a probabilistic approach by modeling aircraft interactions, positions, and velocities as stochastic variables, thereby capturing the inherent uncertainty and variability in air traffic flow. Focusing on lateral movement, another CA [42] modeled aircraft interactions, including dynamic maneuvers, conflict avoidance, separation, and route flexibility under mixed traffic conditions. This CA exhibits phase transitions; for example, at high occupancy, a jammed state emerges, whereas flexible routing creates organized flow zones in the fundamental diagram.

### C. CONTRIBUTIONS OF THIS STUDY

Despite these advances, the existing CA models have several limitations that hinder their applicability to more generalized or operationally diverse contexts. Most importantly, many models rely on rule definitions and airspace representations tailored to specific geographic regions, operational assumptions, or traffic scenarios such as particular airports,

fixed sector layouts, or idealized traffic flows. Consequently, the scalability and transferability of these models to other airspace environments, particularly to those with different structural or procedural characteristics, are limited.

To address this issue, the authors of the present paper studied 1D-CA with step-back movement (SBCA) for arrivals at RJTT [10], [11], [12] and throughout Japan [9]. The unique characteristic of the SBCA is its ability to model air traffic flow with only two parameters and three rules, which is simpler than those used in previous related studies. The flow of air traffic throughout the country is generally modeled using a robust and large-scale scheduling optimization approach [43]. Our simplified modeling enables large-scale simulations with reduced computational complexity while maintaining high fidelity with multiple aircraft behavior. However, it tends to overestimate delays in en-route airspaces where vertical maneuvers such as climbs and descents do not occur. In addition, potential conflicts at the crossing points were not explicitly evaluated in these studies. Based on these findings, the aim of this study was to answer the following research question: *How can a computationally efficient CA model be designed to realistically simulate both delay and conflict dynamics in en-route airspace, accounting for vertical and horizontal aircraft interactions under heterogeneous traffic conditions?*

To this end, this paper proposes a 2D-CA model designed to separately evaluate the delay caused by bottlenecks in air traffic flow and the potential conflicts that occur in hotspot areas. The proposed 2D-CA model addresses the limitations of previous studies by representing aircraft movements along both the direction of the route and the vertical dimension (altitude). The model determines the potential interference based on the separation distance, flight destination, and flight phases, namely climb, cruise, and descent. This design allows for a more realistic representation of traffic dynamics in the airspace. Although conflict resolution is explicitly applied at bottleneck points, potential conflicts at the crossing points between-routes are evaluated using external observations. The interaction between aircraft is selectively applied only when these potential conflicts are likely to result in a delay (e.g., vectoring, holding, speed control), thereby maintaining computational efficiency while improving the fidelity of the delay estimation.

In addition to these methodological advancements, the findings of this study contribute to a broader simulation framework by establishing a scalable and modular architecture. Specifically, the core engine of the simulation was developed and standardized interfaces were defined, by which independently developed simulation parts—or ‘‘bricks’’—can be integrated. Each brick encapsulates a specific operational process and can be reused or rearranged across different spatial and temporal scales, from localized airport taxi operations to nationwide en-route traffic flows. This design not only enables flexible modeling across diverse scenarios but also facilitates rapid prototyping and evaluation of emerging ATM concepts within a unified and computationally lightweight platform.

## III. TWO-DIMENSIONAL CELLULAR AUTOMATON-BASED AIR TRAFFIC SIMULATOR

This section presents the structure and mechanism of the proposed 2D-CA model. The goal was to simulate air traffic dynamics by discretizing the movement of the aircraft in the horizontal and vertical dimensions. This design enables the evaluation of spacing adjustments and potential conflicts under realistic operational constraints.

Fig. 1 illustrates the proposed 2D-CA-based air traffic simulation model. The inputs were derived from various sources, such as CARATS Open Data (COD) [44], the BADA performance database [45], Aeronautical Information Publication (AIP) [46], and standardized flight route repositories. The key flight data parameters integrated into the simulation model include (i) departure and arrival airports or FIR boundary fixes, (ii) runway estimates (from COD and AIP), (iii) reference departure/arrival times, (iv) flight levels, (v) aircraft type (referenced to BADA), and (vi) the nominal flight route determined from the origin-destination pair. Collectively, they define the initial conditions and constraints for the 2D-CA simulation. In addition, the simulation incorporates wind conditions from the mesoscale model grid point value (MSMGPV) data [47] provided by the Japan Meteorological Agency, which include wind direction and wind speed at multiple altitudes. These data were used to calculate the ground speed of each aircraft during the simulations. The MSMGPV data had a temporal resolution of 3 h and a horizontal spatial resolution of 0.125° longitude and 0.1° latitude, with vertical profiles typically available every 25–100 hPa pressure altitude. This design enables the simulation to more realistically capture the influence of atmospheric conditions on aircraft flight movements.

The diagram shown in Fig. 1 visualizes multiple aircraft flying through a series of waypoints in en-route airspace divided into two perspectives: The upper part shows a top-down view highlighting the horizontal (directional) progression, and the lower part provides a side view representing altitude transitions. The different flight phases, including climb (orange), cruise (black), and descent (blue), are distinguished by a color or symbol that reflects the current state of each aircraft. The network allows bidirectional flows, but spacing adjustments are not applied between aircraft traveling in opposite directions along the same waypoint pair, which is analogous to a two-lane road with opposing traffic. Interactions between aircraft occur primarily along converging routes or among aircraft traveling in the same direction. These interactions are calculated using the separation distance and aircraft states, as described in the rules introduced in Section III-D. Each aircraft follows a distinct trajectory governed by the waypoint network and the CA update rules.

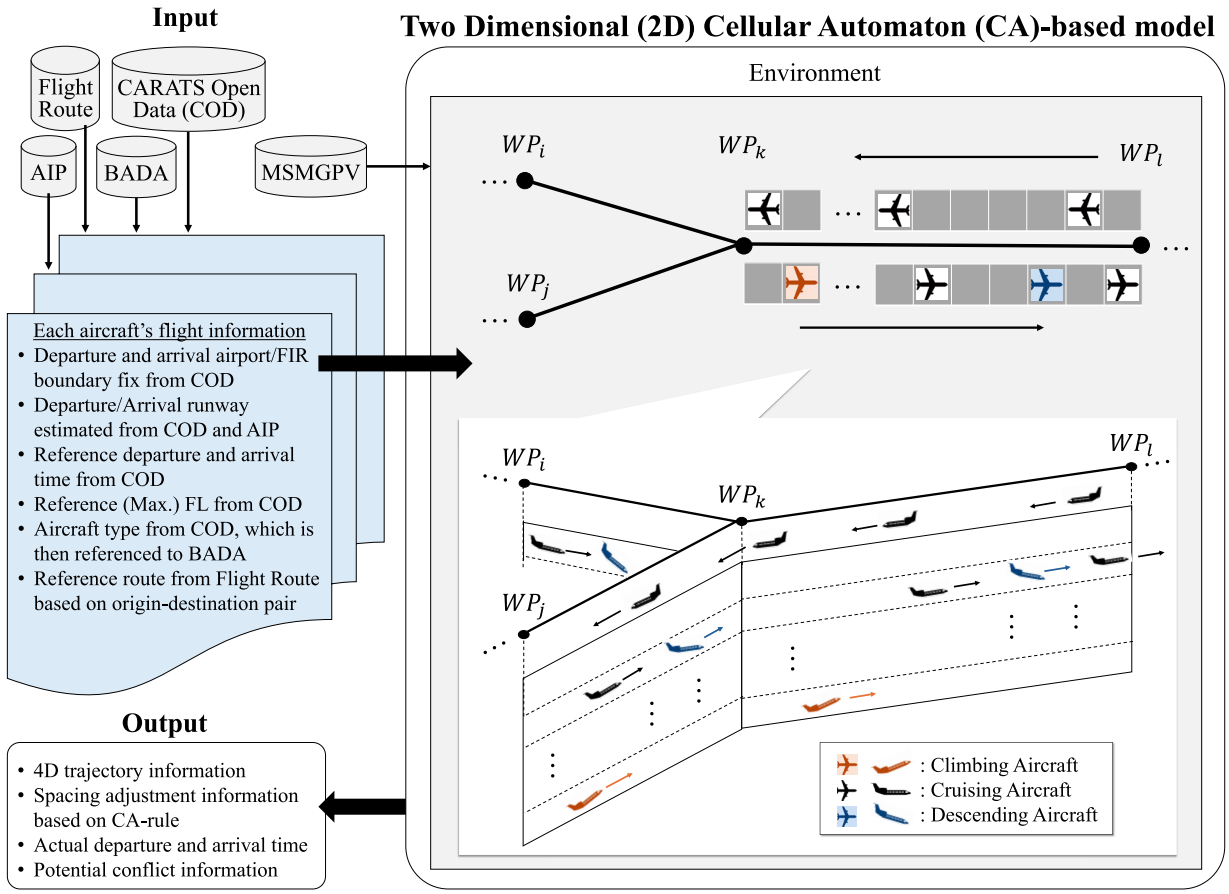



**FIGURE 1.** Overview of the proposed 2D-CA-based air traffic model. This diagram illustrates the inputs, processing flows, and outputs of the model, which uses aircraft flight information, routing data, and environmental conditions to generate 4D trajectories and potential conflicts.

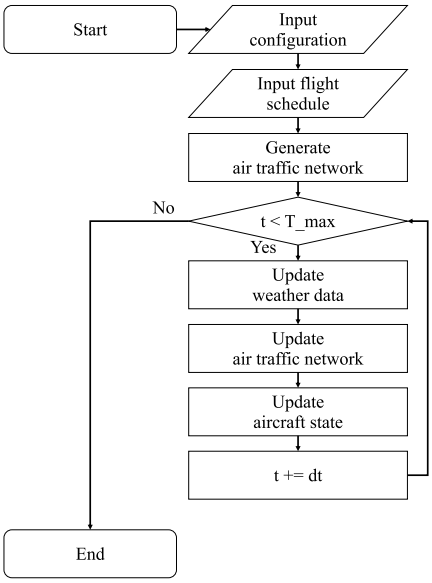



**FIGURE 2.** Flowchart of the time evolution algorithm used in the proposed 2D-CA-based model.

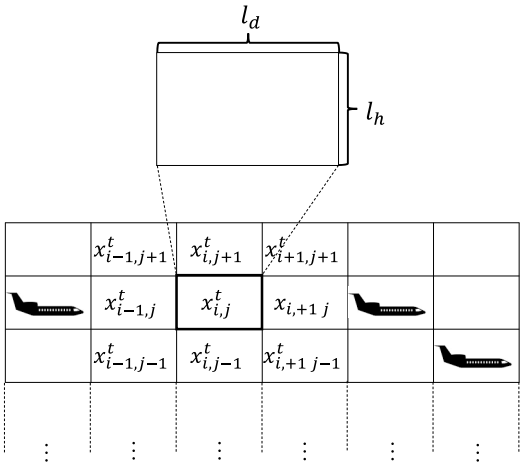



**FIGURE 3.** Pattern diagram of 2D grid neighborhood structure used for the proposed 2D-CA model.

Consequently, 2D-CA generates outputs such as 4D trajectories, spacing adjustment data, actual timing, and potential conflict detections. The term "potential conflicts" denotes latent interactions that were not explicitly addressed by the CA-based spacing-adjustment mechanism because they did not directly influence the separation logic. These interactions were identified through external observations and served as

additional indicators of the emerging complexity within the traffic flow. A detailed discussion of these interactions is presented in Section III-G.

The diagram shown in Fig. 2 illustrates the simulation time-step procedure. Starting from the initial configurations, the model iteratively updates the flight schedules, air-traffic network states, and weather conditions until the maximum simulation time is reached. At each time step, aircraft states evolve according to the CA rules, allowing dynamic interactions and emerging traffic behaviors. This structure supports the efficient forward-in-time simulations of complex airspace scenarios.

### A. SPATIAL STRUCTURE

Fig. 3 shows the local 2D grid structure considered in the 2D-CA model. Each cell $x^t_{i,j}$ represents a binary state variable in the 2D cell, where $x^t_{i,j} = 0$ indicates that the cell is unoccupied, and $x^t_{i,j} = 1$ indicates that aircraft $a$ is present in the cell. The state transitions of the aircraft are determined from the separation distance, and the states of adjacent aircraft are determined as described in Section III-D. The position of each aircraft $a$ in the discretized grid is represented by a pair of indices $(i_a, j_a)$ corresponding to the longitudinal and vertical directions, respectively. The longitudinal index $i_a$ is defined as the floor of the cumulative distance $d_a$ from the source waypoint of the airway segment divided by the cell size $l_d$, whereas the vertical index $j_a$ is defined as the ceiling of the altitude of the aircraft $h_a$ divided by the vertical discretization unit $l_h$.

$$i_a = \left\lfloor \frac{d_a}{l_d} \right\rfloor \tag{1}$$

$$j_a = \left\lceil \frac{h_a}{l_h} \right\rceil \; eq : cell_j \tag{2}$$

where $l_d$ denotes the cell size in the longitudinal direction [m], and $l_h$ is the vertical resolution unit; they were set to $l_d = 70$ m and $l_h = 1000$ ft in this study.

### B. NEIGHBORHOOD

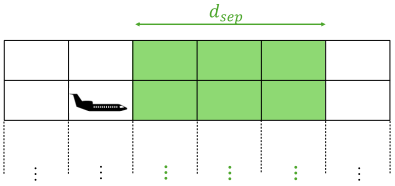



**FIGURE 4. Neighborhood parameter representation. The spatial relationship of the aircraft in the 2D-CA model is governed by a distance-based parameter $d_{sep}$, which defines the degree of assurance of the local neighborhood separation.**

In the proposed 2D-CA model, each cell represents a discrete unit of airspace along both the route (longitudinal direction) and altitude (vertical direction). To maintain safe separation among aircraft and avoid conflicts, a neighborhood parameter based on distance, denoted by $d_{sep}$, was introduced in our previous studies [9], [10], [11], [12]. The parameter defines the longitudinal extent to which aircraft interactions are considered in update decisions.

The neighborhood of an aircraft is defined as a set of cells within a longitudinal range of $d_{sep}$ units ahead of its current position. This range acts as a look-ahead buffer, allowing the aircraft to detect and respond to leading traffic. Fig. 4 illustrates this concept schematically. The value of $d_{sep}$ is critical for ensuring compliance with the required separation standards and depends on the operational airspace. In the en-route airspace, the separation requirement was set to 7 NM and in the terminal areas to 4 NM; the separation distance of the wake turbulence was near the runway threshold. In our model, each cell represents a longitudinal distance of $l_d$, and parameter $d_{sep}$ is defined as the number of cells corresponding to these distances. The neighborhood is dynamically updated at each simulation step according to the current configuration of the aircraft in the grid. Note that the vertical dimension is considered when evaluating the climb/descent transitions.

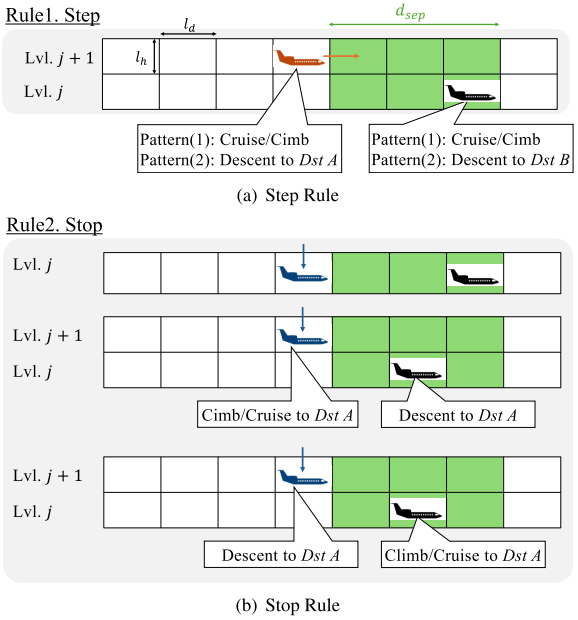



**FIGURE 5. Schematic aircraft movement based on basic rules in the 2D-CA model. Each cell represents a discretized airspace segment along the route (horizontal axis) and altitude level (vertical axis). Solid arrows indicate potential movements governed by the basic update rules, and green grids indicate separation constraints.**

### C. BOUNDARY CONDITION

As shown in Fig. 1, open boundary conditions were used to reflect realistic traffic flow patterns more accurately, with the aircraft entering the simulation domain at one boundary and exiting at another.

### D. BASIC UPDATE RULE

Fig. 5 shows a conceptual diagram of the 2D-CA model. The horizontal axis represents distinct route positions, whereas

the vertical axis corresponds to discretized flight levels in 1,000-ft increments. Each grid cell denotes a potential spatio-temporal slot that an aircraft may occupy.

The aircraft movement on the 2D-CA grid is determined by two behavioral rules, which are applied only to the relevant aircraft and filtered based on the following conditions:

### 1) INTERACTION FILTERING CONDITION

Aircraft $a'$ is considered to interact with the target aircraft $a$ if any of the following holds:

$$\text{(Condition 1)} \quad j_{a'}^{t} = j_{a}^{t}, \tag{3}$$

or

$$\text{(Condition 2)} \quad \left(s_{a}^{t} = \text{descent} \quad \text{or} \quad s_{a'}^{t} = \text{descent}\right) \quad \text{and} \quad \text{dest}_{a} = \text{dest}_{a'}, \tag{4}$$

where $j$ is the discretized flight level index, $s$ denotes the flight state (e.g., cruise, climb, descent) at time step $t$, and dest represents the aircraft arrival airport.

This means that only aircraft satisfying either of the following criteria are considered for the separation checks:

- They are at the same discretized flight level, or
- At least one aircraft is in the *descent* state and both aircraft are arriving at the same airport.

Aircraft in the *cruise* or *climb* phases at different flight levels and heading to different airports are excluded from the interaction. This refinement reflects the operational importance of a descent-phase aircraft converging toward the same destination where potential conflicts are more likely. Consequently, the model more accurately prioritizes potential interactions that are critical for terminal area management and conflict resolution near arrival airports.

### 2) MOVEMENT RULES

- *Step Rule (Forward Movement)*: An aircraft $a$ attempts to move in the longitudinal direction by updating its physical position $(i_a^{t+1}, j_a^{t+1})$. The corresponding target cell indices are computed using (1) and (**??**) as

$$i_a^{t+1} = \left\lfloor \frac{d_a^{t+1}}{l_d} \right\rfloor, \quad j_a^{t+1} = \left\lceil \frac{h_a^{t+1}}{l_h} \right\rceil. \tag{5}$$

  The aircraft moves to the new position $(i_a^{t+1}, j_a^{t+1}) = (i_a^{next}, j_a^{next})$ if the following two conditions are satisfied:

  1) The target cell is unoccupied:

$$x_{i_a^{\text{next}}, j_a^{\text{next}}}^{t} = 0, \tag{6}$$

  2) The longitudinal separation from the preceding aircraft $a'$ satisfies:

$$d_f = i_{a'}^{t} - i_a^{t} \geq d_{\text{sep}}, \tag{7}$$

  where $a'$ is the immediately preceding aircraft on the route at the same flight level or otherwise relevant according to the filtering condition, and $d_{\text{sep}}$ is the minimum longitudinal separation in the cells.

- *Stop Rule (Holding Position)*: If forward movement is blocked by an occupied cell or insufficient separation, the aircraft is in the following position:

$$i_a^{t+1} = i_a^{t}, \quad j_a^{t+1} = j_a^{t}. \tag{8}$$

To ensure consistent update order and conflict resolution, the simulation adopts a breadth-first search (BFS) update mechanism. This method sequentially updates the aircraft positions along the airway graph from downstream to upstream, preventing premature updates that could violate the separation constraints.

**Algorithm 1** Moving Aircraft Mechanism Based on Breadth-First Search (BFS)

**Input:** *InFlightTable* ($ac_i$ for $i = 1$ to $n$); `// Sorted by distance to go (DtG).`
**Input:** $G(route) := (V(waypoint), E(airwaysegment))$
**Output:** All aircraft updated according to BFS order
$t \leftarrow$ Current simulation time;
$num_aircrafts \leftarrow$ Count of aircraft not yet updated;
$depth \leftarrow 0$;
$max_depth \leftarrow$ GetMaxDepth(*InFlightTable*, *Aircrafts*), ;
$num_updated \leftarrow 0$;
**while** $num_updated < num_aircrafts$ **do**
  $local_num_updated \leftarrow 0$;
  **foreach** $ac_i$ *in InFlightTable* **do**
    **if** $ac_i$ *is already updated* **then**
      **continue**;
    $(V_{source}, V_{target}) \leftarrow$ Plan of $ac_i$ at the current depth;
    $E_{current} \leftarrow$ GetEdge($V_{source}, V_{target}$);
    **if** $E_{current}$ *is empty* **then**
      **continue**;
    $n \leftarrow E_{current}$.MoveAircrafts($ac_i, G_{route}$);
    `; // This definition is provided in Algorithm 2.`
    **if** *front aircraft changed and overtaking detected* **then**
      $E_{current}$.MoveAircraftToFront(new front aircraft);
    $local_num_updated \leftarrow local_num_updated + n$;
  $num_updated \leftarrow num_updated + local_num_updated$;
  $depth \leftarrow depth + 1$;

Algorithm 1 introduces a BFS update procedure that prioritizes aircraft movements layer by layer along the route. This structure helps prevent premature updates that would violate the spatial constraints imposed by other types of traffic. The inputs are the route network $G$ and *InFlightTable* sorted by distance to go (DtG), which is the list of aircraft in flight with information on DtG and other attributes. At each iteration depth (representing a sequential segment along the flight path), the algorithm:

1) Identifies the source and target nodes of each aircraft.
2) Retrieves the corresponding airway segment (edge).
3) Applies the update rule (detailed in Algorithm 2) for that segment.
4) Detects overtaking events and adjusts aircraft ordering if necessary.

The BFS approach ensures that aircraft in earlier segments are moved first, thereby allowing a consistent resolution of dependencies due to separation margins.

**Algorithm 2** Moving Aircraft Mechanism on $E_{current}$

```
Input: ac_i, G_route ;                    // From Algorithm 1
Input: AcTable_{E_current} ;              // Aircraft list on current airway segment
Input: d_sep ;                            // Longitudinal separation threshold
Output: num_updated ;                     // Number of updated aircraft
if ac_i ∉ AcTable_{E_current} then
    return 0;
target_table ← GetTargetAircraftTable(ac_i, AcTable_{E_current});
if target_table is empty then
    return 0;
ac_0, E_next ← Get-front aircraft and next edge;
front_margin ← CalculateTargetMargin(ac_0, E_next);
distances ← CalculateLongitudinalDistance(front_margin, target_table);
num_updated ← 0;
foreach ac_j in target_table do
    if ac_j is already updated then
        continue;
    sep_fwd ← distances[j] − d_sep;
    if end of edge is reached then
        Adding ac_j to the end-node order list;
    if sep_fwd > 0 then
        if not at edge end or is first in order then
            Move forward;
        else
            Hold;
    else
        Hold;
    Update lvl (altitude index) of ac_j;
    Update distances;
    Update AcTable_{E_current} ;
    num_updated ← num_updated + 1;
    if ac_j is ac_i then
        break;
Remove aircraft that reached the end;
return num_updated;
```

Algorithm 2 describes how each aircraft evaluates the forward movement in a given airway segment. The update checks for sufficient separation ahead, and determines whether the aircraft can proceed or must hold its position.

- First, it checks whether the current aircraft is registered in the segment table. If not, skip the update.
- Then, it identifies all aircraft ahead in the same segment (*target_table*) and calculates the forward separation (*sep_fwd*) relative to the defined minimum threshold $d_{sep}$.
- If the separation margin is sufficient and the aircraft is not at the end of the edge or first in the update order, it is allowed to move forward; otherwise, it is in position.
- The altitude level of the aircraft is updated if vertical transitions are allowed (for example, climb or descent operations are triggered).
- Aircraft that reach the end of the current segment are queued for entry into the next segment, subject to availability and separation constraints.

This procedure ensures conflict-free updates, adherence to both horizontal and vertical separations in the airway segment, and supports dynamic resequencing when overtaking occurs.

### E. VELOCITY SETTINGS BASED ON AIRCRAFT DYNAMICS

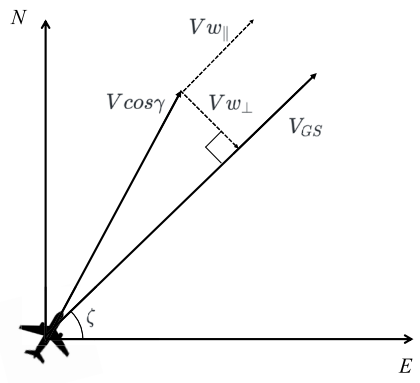



**FIGURE 6.** Conversion from true airspeed (TAS) to ground speed (GS) on the velocities triangle.

To realistically simulate the behavior of an aircraft within the proposed 2D-CA framework, discretized along the lateral (track-aligned) and vertical (altitude) axes, we adopted a point-mass model that incorporates the concept of total energy. This approach allows us to represent aircraft dynamics using reduced-order physical abstraction, while retaining the capacity to account for interactions between individual aircraft based on their energy states. Total-energy modeling is essential in this context for two reasons. First, aircraft interactions in our model are not limited to lateral separation but are significantly influenced by vertical motion, especially when evaluating the potential conflicts described in Section III-G. Second, the model aims to capture operationally relevant variations in the climb, descent, and cruise phases, where altitude profiles and energy management directly influence the ground speed and spacing strategies.

The total energy model relates the aircraft's kinetic and potential energies, allowing for the consistent computation of velocity changes during vertical transitions. Ground speed is a key parameter for determining aircraft position updates and is computed as follows:

$$V_{GS} = \sqrt{(V_{TAS}\cos\gamma)^2 - (V_{w\perp})^2} + V_{w\parallel} \tag{9}$$

Here, $V_{TAS}\cos\gamma$ represents the true airspeed (TAS) projected onto the horizontal plane, considering the flight path angle $\gamma$. The components of the wind velocity parallel and

perpendicular to the aircraft track ($\zeta$) are given by

$$V_{w\parallel} = V_{w_x} \cos\zeta + V_{w_y} \sin\zeta \tag{10}$$
$$V_{w\perp} = -V_{w_x} \sin\zeta + V_{w_y} \cos\zeta \tag{11}$$

The vertical rate of change in the pressure altitude $\dot{Hp}$ represents the rate of rise or descent of the aircraft. Moreover, it is derived from the vertical component of the TAS and is expressed as follows:

$$\dot{Hp} = V_{TAS} \sin\gamma \tag{12}$$

The assumptions included negligible drift angle, vertical wind, and time derivative of wind velocity.

The evolution of an aircraft's motion in the vertical plane is governed by the following set of differential equations derived from the total energy model: Eqs. (13) and (14) describe the time rates of change in the TAS ($\dot{V}_{TAS}$) and flight path angle ($\dot{\gamma}$), respectively.

$$\dot{V}_{TAS} = \frac{Thr - D}{m} - g \sin\gamma \tag{13}$$
$$\dot{\gamma} = \frac{g \cos\gamma}{V_{TAS}} \tag{14}$$

denoted by thrust ($Thr$), which is the force produced by the engines; and drag ($D$), which is the aerodynamic force to counteract an aircraft's movement through the air, aircraft mass ($m$), and gravitational force ($g$). The International Standard Atmosphere (ISA) model was used in this study. In (13), $m$ is not constant but decreases over time because of fuel consumption. The mass-reduction rate was modeled on the basis of a predefined fuel flow rate. The initial mass ($m_0$) is estimated as the weighted average of the operating empty weight ($OEW$) and the maximum takeoff weight ($MTOW$) and is given by

$$m_0 = OEW + 0.7 \cdot (MTOW - OEW) \tag{15}$$

The values of $Thr$ and $D$ were determined by the equations and coefficients given in the BADA3 (Base of Aircraft Data), published by the EUROCONTROL Innovation Hub. In this study, the maximum climb and cruise thrusts ($Thr_{max,climb}$ and $Thr_{max,cruise}$) given by BADA3 were used as the thrust of the aircraft during takeoff and cruise. By contrast, the thrust during the descent phase ($Thr_{des}$) is defined by the altitude and phase.

Fig. 7 illustrates the segmentation of the flight phase used in the simulation based on BADA. Scheduled speed profiles are defined according to standard airline operating procedures and are used to generate altitude-dependent speed schedules across different flight phases (e.g., climb, cruise, and descent). These baseline profiles inform the aircraft state updates during the simulation. In this study, the standard instrument departure (SID) and STAR procedures were not explicitly modeled; therefore, no speed or altitude restrictions associated with such procedures were imposed. During the spacing maneuvers governed by the 2D-CA rules described in (8), the aircraft maintains its last known state with a constant speed, heading, and flight path angle during the adjustment period.

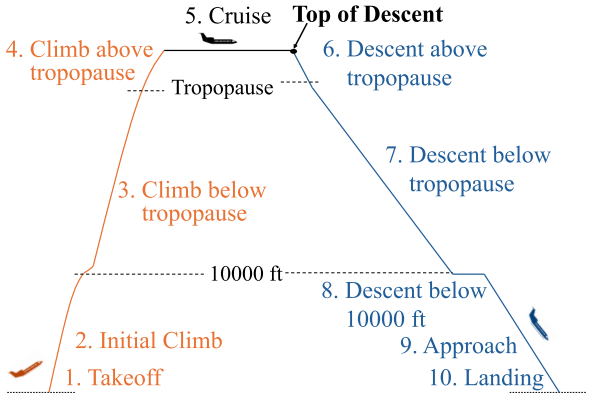



**FIGURE 7.** Flight phase classification for modeling aircraft dynamics in the proposed 2D-CA model.

The Top of Descent (ToD) was computed using the following empirical formula:

$$\text{ToD(NM)} = \frac{\Delta h_{\text{FL}}}{3} + \frac{\Delta V_{\text{TAS(kt)}}}{10} \cdot 2 + \frac{W_{\|(kt)}}{10} \tag{16}$$

Here, $\Delta h_{\text{FL}}$ is the altitude difference between the current cruise level and the target altitude, expressed in flight levels (FL). $\Delta V_{\text{TAS(kt)}}$ denotes the difference between the TAS and the target speed at the runway threshold converted to knots. The target altitude was set as the elevation of the runway, assumed to be 0 ft (= FL0), and the target speed corresponded to the speed of the planned approach at the runway threshold. The ToD is calculated when the aircraft's great-circle distance to the destination falls below 250 NM. The current flight altitude, TAS, and wind information along the track were used. If the aircraft has not yet reached cruise level at 250 NM, the calculation uses the expected cruise performance values (i.e., the TAS and wind conditions anticipated at the planned cruise altitude).

### F. DEPARTURE/ARRIVAL RATE CONTROL BASED ON RUNWAY SEPARATION

Fig. 8 shows how departure and arrival operations are integrated at the grid level. It should be noted that in this simplified airport model, each runway is represented as a single point, and interactions between runways, such as crossings, staggered layouts, or spacing between parallel runways, are not explicitly modeled. The aircraft is aligned with a designated runway threshold and is inserted or removed from the simulation domain on the basis of its proximity to the departure and arrival points. Variables such as takeoff/landing spacing, sequencing (e.g., next departure/arrival), and aircraft intent were considered using Algorithm 3.

In the proposed model, runway availability for departures is managed dynamically to ensure conflict-free operations, as outlined in Algorithm 3. A runway is unavailable for departure under the following conditions.

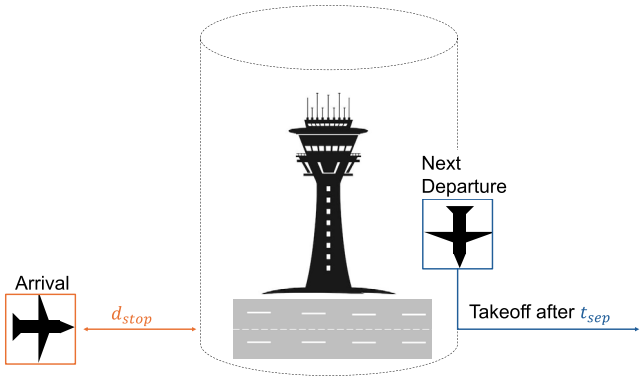



**FIGURE 8.** Schematic illustration of a simple airport model. Each runway in an airport is modeled as a single point that controls the arrival and departure rates. The arrival and departure times were controlled by two key parameters: $d_{stop}$ and $t_{sep}$.. Multiple runways can be represented within the airport brick by modeling each runway independently.

- **Post-departure blocking:** Immediately after a takeoff or landing, the runway is temporarily closed for subsequent departures. The blocking duration for $t_{sep}$ is determined by the wake-turbulence separation time.
- **Arrival proximity blocking:** When an aircraft approaches an airport, the designated landing runway is preemptively marked as unavailable to prevent conflicts. Specifically, if the aircraft DtG is less than a predefined threshold $d_{stop}$, the runway is considered occupied even before landing. $d_{stop}$ was set to 3.5 NM.

This mechanism ensures that no take-off clearance is issued when arrival is imminent, thereby avoiding potential runway incursions. At each simulation time step, the model evaluates whether the next departure time associated with the most recent departure has expired. If the runway is deemed available, a take-off clearance can be granted for the next eligible departure.

### G. CALCULATION OF POTENTIAL CONFLICTS

Fig. 9 illustrates how vertically layered airways are modeled in the 2D-CA framework, with different routes represented as separate lanes distinguished by colors in the diagram. Aircraft following the same route (i.e., on the same-colored lane) interact with one another according to CA rules, such as maintaining separation and resolving conflicts through delay-inducing actions, such as speed control or rerouting. In contrast, aircraft on different routes (i.e., on different colored lanes) do not interact directly under standard CA rules, because conflict resolution in this case is typically handled by altitude changes rather than trajectory extension or speed adjustment, which have a more direct impact on the delay metrics. However, recent studies [7] have shown that potential conflicts between aircraft crossings, particularly on intersecting airways, can significantly increase the controller workload. Therefore, in this simulation model, the potential interactions between the aircraft on different routes were measured externally.

**Algorithm 3** Updating Runway Availability Mechanism at Airports

**Input:** $ac_i$ ; // Aircraft object (approaching flight or takeoff flight)
**Input:** *Airport* ; // Airport object including runways
**Input:** $t_{current}$ ; // Current simulation time
**Input:** $d_{stop}$ ; // Distance threshold to block departures for arriving aircraft
**Output:** None
**if** $ac_i$ *is arriving* **then**
  // Arrival processing
  $DtG \leftarrow$ GetDistanceToGo($ac_i$);
  $runway_end \leftarrow$ GetArrivalRunway($ac_i$);
  $runway \leftarrow$ Airport.GetRunwayFromEnd($runway_end$);
  **if** $DtG < d_{stop}$ **then**
    $runway.is_departure_allowed \leftarrow False$.
  **if** $ac_i$ *has landed* **then**
    // Block departures after landing
    $runway.next_departure_time \leftarrow t_{current} + Airport.arrival_blocking_time$;
    $runway.is_departure_allowed \leftarrow False$;
    $arrival_wake \leftarrow$ GetWakeCategory($ac_i$);
    $runway.previous_wake_category \leftarrow arrival_wake$;
**else**
  // Departure processing
  $runway_end \leftarrow$ GetDepartureRunway($ac_i$);
  $runway \leftarrow$ Airport.GetRunwayFromEnd($runway_end$);
  $runway.is_departure_allowed \leftarrow False$;
  $preceding \leftarrow runway.previous_wake_category$;
  $following \leftarrow$ GetWakeCategory($ac_i$);
  $t_{sep} \leftarrow$ WakeTurbulenceSeparationTime($preceding$, $following$);
  $runway.next_departure_time \leftarrow t_{current} + t_{sep}$;
  $runway.previous_wake_category \leftarrow following$;

Specifically, crossing conflicts were detected on the basis of a threshold of 5 NM for lateral separation and 1,000 ft. for vertical separation. This mechanism allows the quantification of latent conflict levels at intersections, even when direct interactions are not modeled within CA rules.

### H. SIMULATION IMPLEMENTATION AND PARAMETER SETTING

The proposed 2D-CA simulator was implemented using Python. Although the core framework was custom-built, performance-critical elements, such as aircraft performance modeling and trajectory calculations, utilize selected components of pyBADA [48], particularly for file parsing and performance parameter computation.

Table 1 summarizes the major parameters used in the simulations. These include the grid resolution, separation thresholds, aircraft performance references, and runway separation settings. By consolidating these parameters, we aimed to clarify the simulation conditions under which the model was validated.

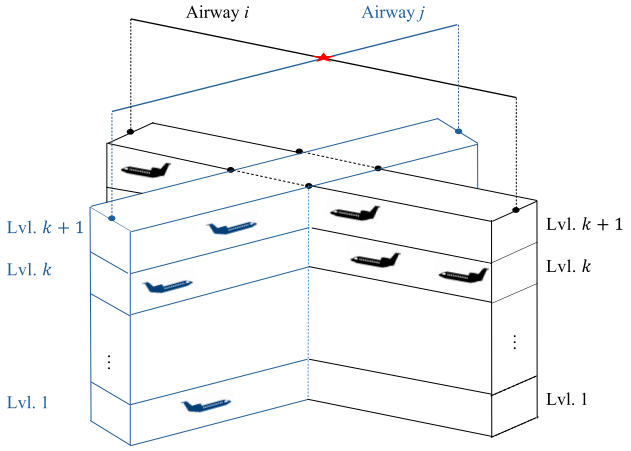



**FIGURE 9.** Schematic diagram of vertical representation of 2D cells that intersect in the airway.

## IV. CASE STUDY I (VALIDATION): ARRIVAL TRAFFIC AT EIGHT MAJOR AIRPORTS IN JAPAN

To validate the proposed 2D-CA model, we analyzed arrival traffic at eight major airports in Japan using COD2019 data. The focus on arrival traffic was motivated by the fact that spacing adjustments such as vectoring and holding are predominantly performed during the arrival phase, particularly in the terminal area near the destination airport. By simulating arrival flows and comparing them with the observed data at these airports, we aimed to assess the model's ability to reproduce realistic traffic dynamics under operational conditions. The selected airports include major international hubs (Haneda, Narita, and Kansai) and key domestic centers (Fukuoka, New Chitose, Naha, Itami, and Chubu Centrair), which offer a diverse set of operational environments in terms of traffic volume, complexity, and regional characteristics.

To facilitate validation, we adopted a multifaceted comparison framework comprising qualitative and quantitative indicators. First, we qualitatively compared the actual flight tracks with the trajectories generated by the proposed 2D-CA model, with a focus on identifying the regions where spacing adjustments were applied. In particular, we examined whether the model spacing actions, which were derived from the 2D-CA rules, aligned with the actual vectoring maneuvers observed in the data. The extension of the model to the vertical dimension also allowed the visualization of overlapping aircraft paths in a two-dimensional space, providing additional information on the separation logic.

Second, we quantitatively evaluated the number of spacing adjustments in both the airspace en-route (250–80 NM from the airport) and at the terminal (80–0 NM). This test enabled us to assess whether the model reproduced realistic adjustment frequencies within operationally plausible timeframes. Finally, we compared the total flight distances in the en-route and terminal phases between the simulated and actual trajectories. This metric captures the effects of path extensions and detours, thereby reflecting whether the model realistically accounts for typical arrival-flow management practices. Through these three layers of analysis (qualitative pattern matching, adjustment frequency statistics, and flight-distance comparisons), our objective was to assess the extent to which the proposed 2D-CA model could replicate operational arrival traffic characteristics.

**TABLE 1.** Key parameter settings used in the 2D-CA simulation. These parameters apply to all experiments unless otherwise stated.

| Parameter | Value / Source | Description |
|---|---|---|
| $l_d$ | 70 m | Grid resolution along the flight route |
| $l_h$ | 1,000 ft | Grid resolution in altitude |
| $d_{\text{sep}}$ | 7 NM (en-route), 4 NM (TMA), Wake turbulence distance separation (runway) | Horizontal separation threshold based on airspace type |
| $t_{\text{sep}}$ | 60–180 s (BADA-based) | Time separation at the runway threshold based on wake turbulence |
| $d_{\text{stop}}$ | 3.5 NM | Distance threshold to block departures for arriving aircraft |
| $\Delta t$ | 1 s | Update interval for aircraft state transitions |
| $m_0$ | pyBADA (aircraft-specific) | Takeoff weight for each aircraft calculated using (15) |
| Aircraft performance data | pyBADA | Used for TAS, thrust, drag, mass, and ToD calculation |
| Wind data | MSM-GPV | 3-hourly wind fields with 0.1 to 0.125° horizontal and 25 to 100 hPa pressure-altitude vertical resolution |
| ToD location | pyBADA and MSMGPV-based estimation | Calculated using (16) with altitude, TAS, and tail/heading wind |

### A. DATA DESCRIPTION

Table 2 presents a statistical summary of the daily aircraft arrival counts, showing the median ($\tilde{N}_{ac}$) and quartile deviation (QD $= (Q_3 - Q_1)/2$) calculated from 84 days of data, covering one representative week per month throughout 2019. In addition, the table includes the number of operational runways at each airport as of 2019. Some airports such as Fukuoka and Naha have added new runways; however, the configuration used in the table reflects the situation during the observation period.

Using the arrival traffic data listed in Table 2, we constructed input files for the simulation. To perform the simulation, we first selected 10 representative days from COD2019 for each airport based on the runway configuration and prevailing wind conditions (e.g., north- or south-wind operations). Each selected day corresponded to a period of stable runway use, ensuring that the airport operated under consistent arrival procedures throughout the day. Details of the basic parameter settings and structure of the input files can be found in Section III, particularly in the system overview

**TABLE 2.** Median ($\tilde{N}_{ac}$) and quartile deviation ($QD = (Q_3 - Q_1)/2$) of daily aircraft arrival counts at eight major Japanese airports, based on 84 days of COD 2019 data (one week per month). The number of operational runways is also shown for each airport.

| Airport (ICAO / IATA) | Runways | $\tilde{N}_{ac}$ | $QD$ |
|---|---|---|---|
| Haneda (RJTT / HND) | 4 | 615.57 | 3.88 |
| Narita (RJAA / NRT) | 2 | 354.93 | 3.93 |
| Kansai (RJBB / KIX) | 2 | 278.36 | 5.45 |
| Fukuoka (RJFF / FUK) | 1 | 234.29 | 3.12 |
| New Chitose (RJCC / CTS) | 2 | 203.14 | 12.08 |
| Naha (ROAH / OKA) | 1 | 177.21 | 1.63 |
| Itami (RJOO / ITM) | 2 | 173.57 | 3.98 |
| Chubu Centrair (RJGG / NGO) | 1 | 153.07 | 4.68 |

shown in Fig. 1. Because of limited data availability, we had access to flight plans for only a single representative day in 2019. Therefore, the routes on that day were used as representative trajectories. For each origin–destination (OD) pair, a unique route was assumed on the basis of available flight-plan information. This simplification was considered acceptable, because there is generally little variation in the routing structure for a given OD pair under normal operations, particularly in Japan's highly structured airspace. Thus, its impact on the simulation results was expected to be minimal. Furthermore, each flight plan included information on the waypoint to the terminal maneuvering area (TMA) entry point along with the destination airport. However, because the final approach segment was not explicitly included in the original data, we inserted a notional final approach fix (FAF) extracted from the AIP to complete the route. This step ensured that aircraft approaching from different directions but landing on the same runway could be merged appropriately near the runway threshold.

### B. RULE EXECUTION VALIDATION

Fig. 10 shows an overlay of the execution points of the rules of the 2D-CA model and the actual flight trajectories of COD2019. The results demonstrate a clear spatial correspondence between the locations where the model executed the separation rules (red points) and the aircraft appeared to perform vectoring or spacing maneuvers in actual operations. In particular, the consistent clustering of rule execution points near the convergence areas of arrival flows, such as within 50 NM of RJTT and RJAA, indicates that the model accurately captured typical congestion patterns. Furthermore, the presence of horizontal overlapping points (green), particularly in the southwestern part of the RJTT arrival, reflects the appropriate handling of the vertical separation logic of the model, where interactions at different altitudes are excluded by the interaction filtering condition. These findings suggest that the proposed model provides a realistic representation of traffic and controller interventions, thereby providing a reliable foundation for further quantitative validation.

Table 3 summarizes the statistics of the additional time induced by the spacing maneuvers in the CA simulation over a 10-day period. Among the airports analyzed, RJTT and RJAA exhibited the highest number of spacing events, particularly in the 80–0 NM segment, with average daily flight counts of 314.4 and 128.9, corresponding to 51.07% and 36.3% of $\tilde{N}_{ac}$ in Table 2, respectively. This result reflects both the high arrival demand and the complexity of managing converging flows in the Tokyo Approach Control Area. RJFF shows a notable case with the highest average spacing time in the 80–0 NM segment at 101.6 s, although it has a lower traffic volume than RJTT or RJAA. This value is particularly striking because it aligns closely with the required runway separation time for the arriving aircraft, suggesting that, on average, each aircraft experiences a delay equivalent to waiting for one preceding aircraft to land. One likely contributing factor is that among the airports analyzed, RJFF had the highest number of takeoffs and landings per runway, resulting in intense competition for runway access and tighter sequencing requirements. In general, these results support the expectation that spacing interventions intensify as aircraft approach the airport and that both the traffic volume and structure of the local airspace strongly influence the magnitude and variability of the separation control duration.

**TABLE 3.** Spacing time due to separation control rule executed in the CA simulation. The 250–80 NM segments and 80–0 NM segments nearly correspond to en-route and terminal airspace.

| Arrival Airport | Distance Ring | Flight (avg/day) | Mean (sec) | SD (sec) | Max (sec) |
|---|---|---|---|---|---|
| RJTT | 250–80 NM | 86.5 | 43.1 | 31.3 | 176 |
| | 80–0 NM | 314.4 | 95.3 | 82.2 | 769 |
| RJAA | 250–80 NM | 42.9 | 44.3 | 33.5 | 278 |
| | 80–0 NM | 128.9 | 87.5 | 90.3 | 1067 |
| RJBB | 250–80 NM | 15.7 | 39.2 | 28.6 | 162 |
| | 80–0 NM | 107.0 | 91.5 | 83.0 | 761 |
| RJFF | 250–80 NM | 15.9 | 50.8 | 41.5 | 331 |
| | 80–0 NM | 102.0 | 101.6 | 95.5 | 922 |
| RJCC | 250–80 NM | 45.1 | 48.9 | 41.2 | 258 |
| | 80–0 NM | 75.6 | 83.5 | 65.0 | 486 |
| ROAH | 250–80 NM | 22.5 | 47.6 | 31.9 | 159 |
| | 80–0 NM | 56.7 | 70.8 | 59.7 | 353 |
| RJOO | 250–80 NM | 11.6 | 42.8 | 37.4 | 246 |
| | 80–0 NM | 64.0 | 96.7 | 94.1 | 770 |
| RJGG | 250–80 NM | 5.3 | 47.6 | 46.8 | 271 |
| | 80–0 NM | 34.4 | 73.3 | 58.8 | 347 |

### C. QUANTITATIVE COMPARISON BETWEEN SIMULATION AND ACTUAL OPERATIONAL DATA

Fig. 11 presents a comparison of the flight distance distributions between the proposed 2D-CA model and the actual flight trajectories derived from COD2019, stratified by arrival airport (e.g., RJTT and RJAA). The overall shape of the distributions produced by the CA model closely resembled those observed in the real-world data for most airports. This result suggests that the model captures the key structural characteristics of the actual flight operations. The locations of the distribution peaks and spread (i.e., variability) in the CA model often aligned well with those in the COD2019 data, indicating a reasonable reproduction of typical flight distances. Nonetheless, the peak distributions, especially for the 80–0 NM segment, tended to be underestimated. This

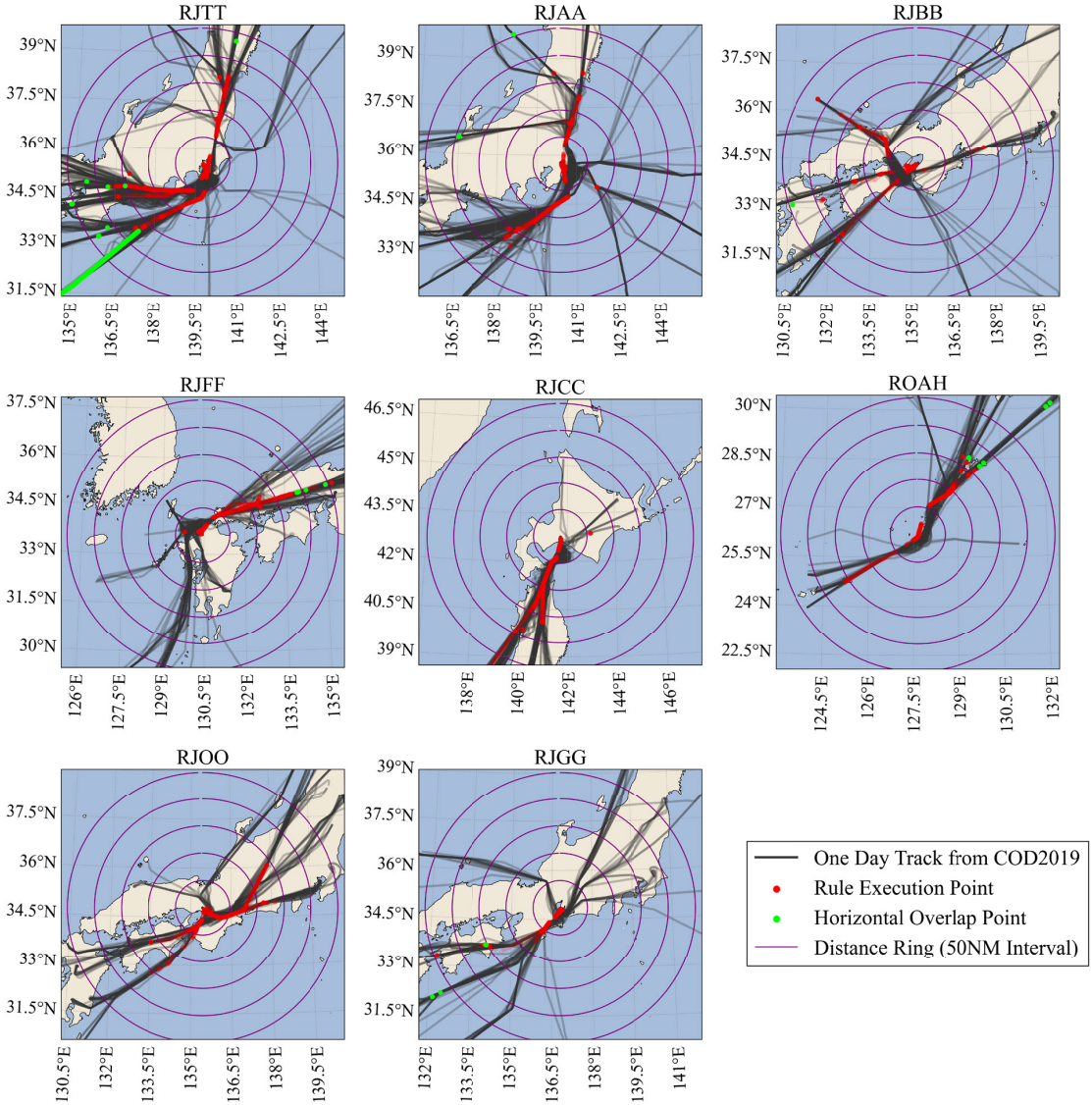



**FIGURE 10.** Actual trajectory in a day from COD 2019 with the rule execution points and horizontal overlap points in the proposed 2D-CA model. The black line (COD2019 One-Day Track) shows the actual track of the arrival traffic flow at the airport on a single day extracted from COD2019. The purple line (Distance Ring (50-NM) Interval) shows the concentric lines from the arrival airport. The red dots (Rule Execution Points) indicate the locations at which the aircraft executed the rules of the CA model (particularly the Stop Rule) to maintain an appropriate interval with the other aircraft. The green dots (Horizontal Overlap Points) indicate the intersection points of the aircraft flying at different altitudes at the same horizontal position. This approach allowed us to check overtaking actions that are determined not to interfere, owing to differences in altitude.

discrepancy can be attributed to the simplified assumptions of the model, such as

1) SIDs and STARs were not explicitly modeled.
2) Horizontal maneuvers, such as holding patterns or vectoring, were not represented in detail and were captured only through simplified stop rules.
3) The takeoff and landing behaviors were governed by simplified rules, without considering detailed runway configurations or sequencing strategy.

However, for all the arrival airports analyzed, there was no clear indication of systematic overestimation or underestimation of the simulated flight distances, which supports the general robustness and transferability of the model. These qualitative results demonstrate that the proposed CA model can approximate realistic flight distances and capture major structural patterns.

## V. CASE STUDY II (GLOBAL ANALYSIS): AIR TRAFFIC IN THE FUKUOKA FIR IN JAPAN

To demonstrate the applicability of the proposed 2D-CA model to large-scale traffic analysis, we simulated a full day of air traffic within the Fukuoka FIR, covering 4,666 aircraft operations. Unlike the previous validation-focused case

study, the aim of this analysis was to provide system-wide insights into traffic complexity and sector-level constraints.

The primary objective was to identify the sectors that act as bottlenecks and potentially hazardous areas under high-density traffic conditions. For this purpose, we focused on the upper airspace (flight levels at or above FL335), where en-route traffic is the densest and separation assurance is critical.

We performed quantitative and visual analyses of potential conflicts and the separation of control activities across the FIR. Potential conflict and separation control are defined as follows:

- **Potential conflict**: Aircraft on **different routes** in proximity (5 horizontal NM, 1, 000 vertical ft)
- **Separation control**: Interactions between aircraft on the **same route**, resolved through spacing actions in the 2D-CA

Subsequently, we identified hotspot sectors in which conflict concentrations or excessive control actions were observed. For these sectors, we further examined detailed aircraft interaction patterns to highlight the key contributors to local traffic complexity. This case study demonstrated the utility of the model for simulating realistic trajectories and enabling large-scale conflict detection, sector-workload assessment, and strategic airspace planning.

### A. DATA DESCRIPTION

The input data used in this analysis consisted of the flight plan information for a full day of operation within the Fukuoka FIR, comprising 4,666 aircraft on a specific day in 2019. These flight plans served as baseline inputs for the 2D-CA simulation. Specifically, the planned departure and arrival times, along with cruise altitudes and waypoints, were directly used to initialize the simulation, reflecting the intended trajectories under normal operating conditions without tactical interventions.

All aircraft operating within the Japanese airspace on the selected day were included to ensure comprehensive coverage of nationwide air traffic. Its scope includes domestic and international flights traversing the Fukuoka FIR.

Fig. 12 shows the flight routes and airspace sectors as a network, where the nodes represent the navigation points and the edges represent the planned flight paths. The network airways are shown in gray, and the main airports (departure/arrival nodes), such as RJTT and RJAA, are marked in red. The resulting topology reveals distinct traffic corridors, which are especially dense in the western part of the area, reflecting the high concentration of flows converging on the main hubs (RJTT/RJAA), as shown in Fig. 12 (140°E, 35°N).

### B. POTENTIAL CONFLICT AND SEPARATION CONTROL IN UPPER AIRSPACE

Fig. 13 and Table 4 present the number of potential conflict flights, which are aircraft flying in different directions with possible proximity issues, and separation control flights, which fly in the same direction and were performed using CA rules to maintain safe intervals for each sector in the upper airspace of Japan with FL335 and above. The highest number of potential conflict flights was observed in sector T25 (52 flights), followed by sectors F09 (31), F07 (27), and F16 (27). These regions are likely to represent airspaces with intersecting flows or major traffic convergence points, highlighting the importance of multidirectional conflict strategies, beyond controlling the same heading. The number of separation control flights was highest in sectors F14 and F12 (22 each), followed by T14 (20), F10 (15), and T25 (15), indicating high-density flows along the same route with the same FL or different aircraft states in these areas. The total delay time owing to the separation control rule observed in this simulation was 1,050 s in sector F14, which is approximately 3.5% of the total number of separation control actions estimated within 80 NM of RJTT, as derived from Table 3 (314.4 flights per day $\times$95.3 actions per flight $=$ 29, 962.32 per day). This comparison illustrates that, although the model captures significant control dynamics at the en-route level, the overall scale remains modest relative to the terminal area.

Fig. 14 illustrates the spatial distribution of the potential conflict points identified in Sector T25, which was found to have the highest number of potential conflicts among all the sectors in the nationwide simulation. As highlighted by the blue arrows, T25 is characterized by a significant crossing of eastbound departures from RJTT/AA and arrivals to RJTT from the west-southwest [49]. In this sector, a notable concentration of conflicting flights was observed, with 27 RJTT/AA departures and 16 RJTT arrivals, which can be attributed to several operational and structural factors. Because eastbound and westbound flights are generally separated using odd and even flight levels, respectively, conflicts are unlikely to occur during level cruises. However, in this region, RJTT arrivals typically initiate descent, whereas RJTT/AA departures have not yet reached the top of climb, resulting in overlapping altitude profiles and an increased potential for conflicts. Furthermore, both RJTT and RJAA serve as the main hub airports, and, as shown in Table 2, the volume of traffic associated with these airports is considerably high. T25 functions as a convergence point for these high-density flows, effectively creating a structural chokepoint. Consequently, T25 exhibited the highest number of potential conflicts in the nationwide simulation.

## VI. DISCUSSION

Before discussing our results, we first restate the main objective of this study: to develop a novel 2D-CA model for air traffic simulation that explicitly represents horizontal and vertical aircraft movements in airspace en-route. The model was designed to evaluate the delays induced by traffic bottlenecks and identify potential conflict regions while maintaining computational efficiency suitable for large-scale applications. The key features of the proposed model are as follows.

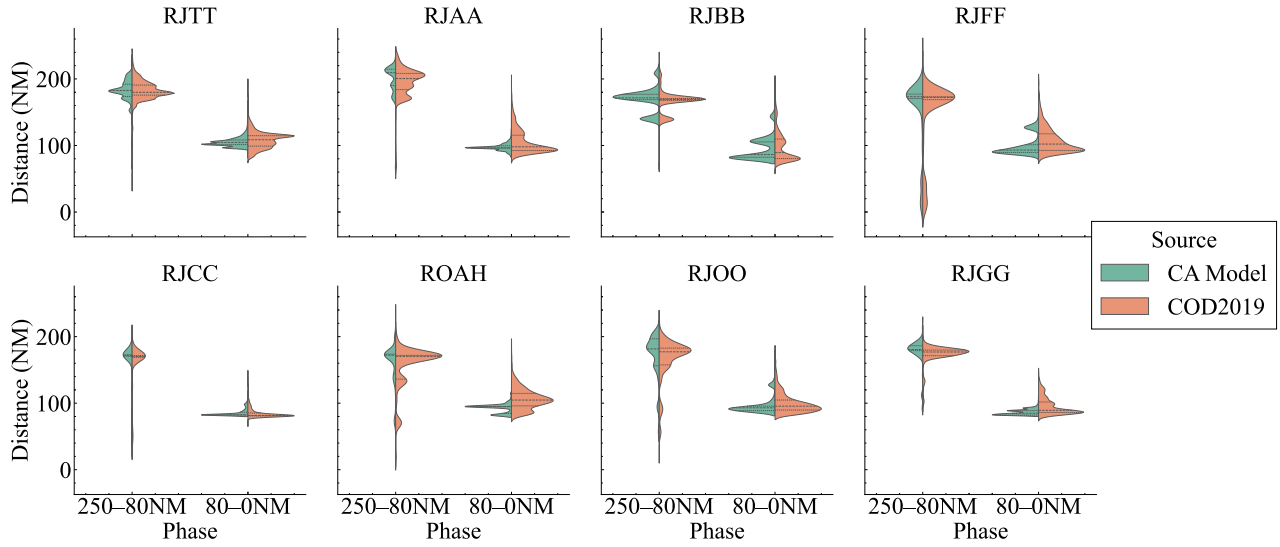



**FIGURE 11.** Comparison of flight distance distributions between the proposed CA model and actual trajectory data (COD2019), grouped by arrival airport. The distributions are plotted with the dashed vertical lines indicating the first quartile (Q1), median (Q2), and third quartile (Q3), providing a summary of the central tendency and spread.

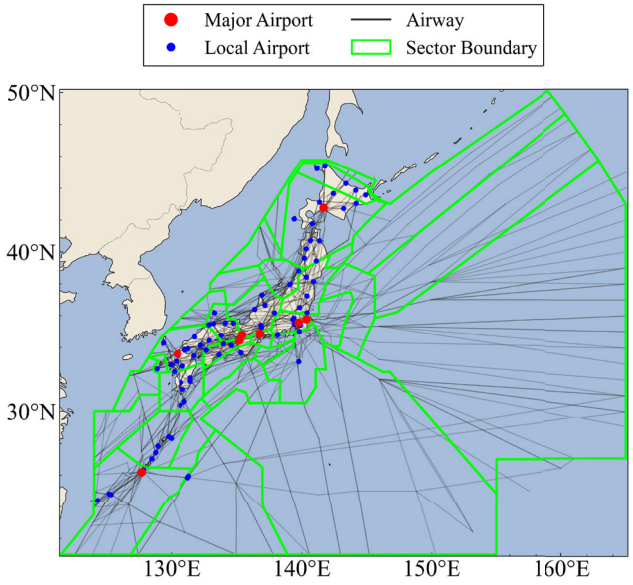



**FIGURE 12.** Overview of planned flight routes and airspace structure in Fukuoka FIR. The black lines represent flight routes based on flight plans, red dots indicate eight major airports, blue dots indicate 81 local airports, and green lines denote the sector boundaries. This visualization highlights the major traffic corridors and their convergence around high-density regions, such as the western area.

**TABLE 4.** Summary of potential conflict flights (between aircraft on different airways) and separation control flights (adjustments between aircraft on the same airway) for each airspace sector.

| Sector | Potential Conflict Flight | Separation Control Flight |
|---|---|---|
| A01 | 0 | 1 |
| A02 | 2 | 1 |
| A03 | 2 | 1 |
| A04 | 0 | 0 |
| A05 | 4 | 0 |
| F01 | 4 | 7 |
| F07 | 27 | 14 |
| F08 | 10 | 13 |
| F09 | 31 | 9 |
| F10 | 23 | 15 |
| F11 | 4 | 8 |
| F12 | 8 | 22 |
| F13 | 6 | 1 |
| F14 | 25 | 22 |
| F15 | 10 | 14 |
| F16 | 27 | 14 |
| F17 | 8 | 0 |
| F94 | 0 | 0 |
| T01 | 12 | 2 |
| T09 | 0 | 0 |
| T10 | 2 | 12 |
| T14 | 12 | 20 |
| T25 | 52 | 15 |
| T92 | 18 | 9 |
| T93 | 4 | 5 |
| T95 | 4 | 2 |

1) The model discretizes both the longitudinal (route direction) and vertical (altitude) axes, enabling the tracking of the climb, cruise, and descent phases with phase-specific interaction rules. These features are crucial for conflict analysis in altitude transition zones.
2) Aircraft interactions (e.g., spacing adjustments such as vectoring or holding) are triggered only when operationally necessary, thereby preserving computational tractability while capturing key delay mechanisms.
3) Potential conflicts at airway intersections are identified on the basis of spatial proximity criteria, even when they do not result in immediate delays, allowing hotspot detection without full-resolution control logic.

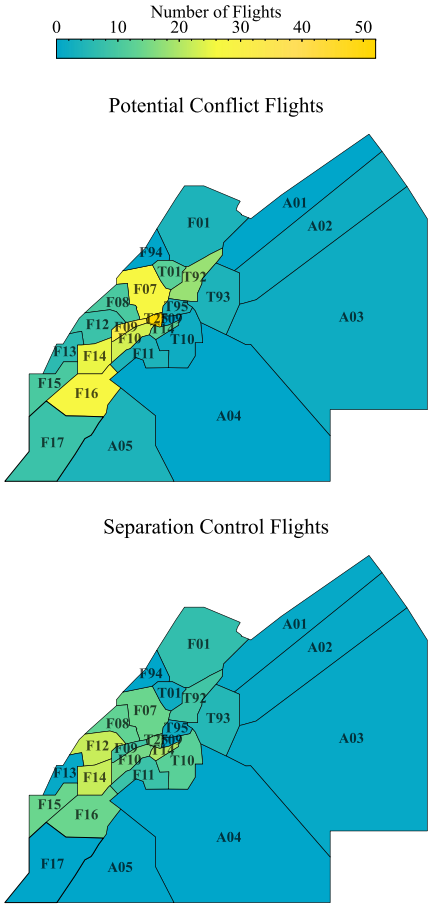



**FIGURE 13.** Sector-wise distribution of potential conflict flights and separation control flights observed in the national simulation of the Japanese airspace. Potential conflict flights represent aircraft flying on different airways with a spatial proximity risk, whereas separation control flights indicate the number of aircraft on the same airway that require interval adjustments. These results illustrate the spatial patterns of the traffic complexity and control demand captured by the proposed 2D-CA model.

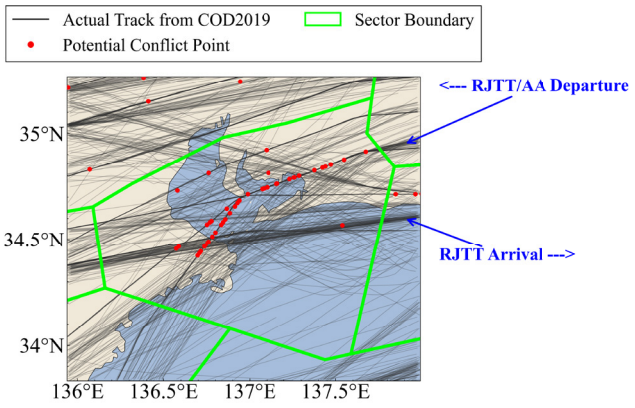



**FIGURE 14.** Visualization of potential conflict points in Sector T25 based on a nationwide air traffic simulation using the proposed 2D-CA model. The black lines represent the actual flight tracks from the COD2019 dataset, the red dots indicate potential conflict points between aircraft flying in different directions, and the green lines denote the sector boundaries. The blue arrows highlight representative conflicting traffic flows: RJTT/AA departures heading southwest and RJTT arrivals from west to northwest.

These features allow the proposed 2D-CA model to bridge the gap between coarse-grained flow models and computationally intensive agent-based simulations, offering a scalable and operationally relevant alternative to air traffic congestion and complexity analyses.

A key strength of the model is its ability to reproduce realistic spacing behaviors and trajectory lengths, as validated through a comparative analysis with operational data from eight major airports in Japan. In particular, the model successfully captured the characteristic spacing adjustments near congested terminal areas (see Fig. 10 and Table 3) and demonstrated strong agreement with the actual flight distance patterns (see Fig. 11). Furthermore, a nationwide simulation covering the Fukuoka FIR revealed the ability of the model to detect hotspot sectors at FL335 and above, such as sectors T25 and F14 (Fig. 13). Sector T25 exhibited a high density of cross-traffic and conflict (see Fig. 14), primarily driven by 16 descending arrivals from RJTT and 27 climbing departures from RJTT and RJAA, which intersect in this region. In such scenarios, ATCos are forced to delay their arrival descent or restrict their departure climb to maintain vertical separation while metering the RJTT arrivals, thereby significantly increasing their task volumes. By contrast, Sector F14 had the highest number of control events with the same direction of separation. These typically involve speed control, minor altitude adjustments, or vectoring during descent, which contribute to the metering complexity for the ATCos. However, because F14 accounts for only a small number of delays compared with the terminal airspace (approximately 3%), its overall impact on the operational task volume is limited. Considering that handling intersecting traffic flows from different directions imposes a higher workload on the controllers [7], the task volume in T25 is likely to be more operationally significant than that in F14, despite the larger volume of spacing interventions in the latter.

Compared with existing approaches, the proposed model achieves a favorable balance between generality, computational tractability, and operational relevance. The model is not bound to a specific terminal [36], [40], [41], [42] or en-route airspace [38], thereby enabling simulations across different airspace configurations with minimal customization. The incorporation of altitude transitions and phase-dependent interaction rules supports the simultaneous assessment of delay and complexity. Furthermore, the proposed model avoids unnecessary spacing actions resulting from route overlaps and allows altitude-based overtaking.

The proposed model is an advance over earlier 1D-CA models [9], [11], [12], [35], [39], which tended to overestimate en-route delays owing to limited vertical maneuvering. A previous study using 1D-CA in the same FIR estimated a total of 333,364 spacing adjustments per day [9], whereas the current two-dimensional model reported 129,853 (Section V's result without FL filtering), a reduction of over

60%. This result suggests that the previous approach significantly overestimates delay-inducing interactions, largely because of its inability to resolve vertical separation, as argued in [9]. As shown in the RJTT panel of Fig. 10, the multiple green points on a specific airway indicate overlapping trajectories throughout various corridors in Section V's simulation in the Fukuoka FIR.

Nonetheless, several limitations that may have contributed to the simulation errors must be acknowledged. The current implementation omits the explicit modeling of SID and STAR, including the associated speed and altitude constraints, which were considered in related studies [40]. This simplification reduces the preparation workload, but can lead to an underestimation of the trajectory length and minor inaccuracies in conflict altitude detection in en-route airspaces. Thus, lateral control measures, such as the heading changes modeled in [42], were not explicitly modeled; instead, their effects were implicitly represented through spacing adjustment rules within the 2D-CA framework. The 2D nature of the proposed model inherently limits its ability to capture vertical maneuvers with full fidelity, such as level-off procedures, altitude changes under ATC instructions, or complex three-dimensional separation strategies. While the discretized vertical layers provide a reasonable approximation for large-scale traffic dynamics, further extensions toward a full 3D-CA framework would be necessary to capture detailed vertical interactions and to better scale the model to ultra-dense traffic environments. Furthermore, airports and runways, modeled in [30], [31], [32], and [33], were also simplified as departure-arrival points by assuming a first-come, first-served, arrival-preferential scheduling mechanism without incorporating constraints related to runway configuration or assignment. In this simulation, the runway used for each arrival and departure was extracted from the actual operational data to approximate realistic sequencing behavior. In today's operations, both runway assignment and the sequencing of arrivals and departures are tactically determined by approach controllers and vary significantly across airports, making general rule formalization inherently difficult. Similarly, the ToD estimate did not account for traffic-induced descent adjustments, in which ATCos frequently instruct pilots to descend in the earlier arrival stage in a congested airspace. The timing of such interventions is also difficult to generalize, and was therefore not included as a rule in this study. Finally, the spatial resolution in the current simulation was set to 70 m in the longitudinal direction (approximately equivalent to the length of an aircraft) and 1,000 ft in the vertical dimension to provide the lowest computational cost with acceptable accuracy. Although increasing the resolution could improve the fidelity of climb/descent dynamics and separation modeling, it would also substantially increase the computational cost.

In addition to our progress in methodology, the proposed model provides a flexible simulation framework that is both scalable and modular. We developed a core simulation engine that accommodated the entire Japanese airspace, demonstrating the feasibility of integrating all functional components—or "bricks"—into a nationwide model. In addition, the framework allows decomposition into minimal operational patterns tailored to specific airspace structures, such as runway interactions at departure and arrival airports, en-route sectors with crossing points, and terminal areas with multiple merging flows. The configuration of each brick can be easily customized by adjusting parameters, such as route length, aircraft appearance times and locations, and separation thresholds. This design enables rapid prototyping and localized scenario testing without rewriting the entire simulation logic. Future work will focus on independently validating each minimal brick, with the goal of creating an open library of reusable simulation modules. Although the primary focus of the present study was en-route operations, the *Aerospace Bricks* framework is also intended to be extended to surface operations, which will be explored in subsequent phases of this project. Another promising avenue is the integration of the CA-based framework with machine learning modules. For example, data-driven predictors could provide short-term forecasts of sector demand or delay propagation, which, in turn, could be injected as dynamic inputs to the simulation. Similarly, reinforcement learning or supervised learning methods could be coupled with conflict resolution mechanisms, allowing the system to evaluate alternative resolution strategies under varying traffic conditions. This hybrid approach would combine the scalability and transparency of the mechanistic CA model with the adaptability of machine learning, thus extending the relevance of the framework to both research and operational decision support.

## VII. CONCLUDING REMARKS

In this study, a 2D-CA model was introduced that realistically represents the dynamics of horizontal and vertical aircrafts in an airspace in the air. The model enables the simultaneous evaluation of delays caused by traffic bottlenecks and potential conflicts arising at crossing points by selectively applying interaction rules based on aircraft phase and proximity. Validation against actual 10-day traffic at eight major airports demonstrated the model's ability to reproduce typical spacing adjustments and realistic trajectory lengths, with spacing delays per flight ranging up to 95.3 s within 80 NM of RJTT. A nationwide simulation also revealed high-complexity upper sectors at FL335 and above, such as T25, where 52 aircraft participated in altitude-overlapping crossings and sectors F12 and F14, which experienced 22 spacing adjustments in the same direction.

Looking ahead, the model provides a flexible and scalable platform for evaluating strategic initiatives, including conflict resolution and procedural modifications, in the diverse operational settings specified by the user. Incorporating more detailed procedures (e.g., SID/STAR constraints and runway operation) and expanding the resolution to a higher spatial resolution could further enhance fidelity. By bridging the

gap between coarse-flow models and detailed agent-based simulations, the proposed model lays the foundation for a scalable data-driven airspace design and traffic management analysis.

## ACKNOWLEDGMENT

The air traffic flow simulator used for the experiment was developed by MEXT as "Program for Promoting Research on the Fugaku Supercomputer." The authors would like to thank Editage (www.editage.jp) for English-language editing.

## REFERENCES

[1] International Air Transport Association (IATA). (2024). *Fact Sheet: Industry Statistics*. Accessed: Aug. 29, 2024. [Online]. Available: https://www.iata.org/en/iata-repository/pressroom/fact-sheets/industry-statistics/

[2] International Air Transport Association (IATA). (2023). *Global Outlook for Air Transport: A Local Sweet Spot*. Accessed: Aug. 29, 2024. [Online]. Available: https://www.iata.org/en/iata-repository/publications/economic-reports/global-outlook-for-air-transport—december-2023—report/

[3] EUROCONTROL and FAA. (Jan. 2024). *Comparison of Air Traffic Management Related Operational and Economic Performance: U.S.–Europe*. [Online]. Available: https://transport.ec.europa.eu/document/download/31ad10d4-51e3-43cf-8592-edbfb69f7ff0en?filename=2024-01-USEurope-comparison-ANSperformance.pdf

[4] J. Djokic, B. Lorenz, and H. Fricke, "Air traffic control complexity as workload driver," *Transp. Res. C, Emerg. Technol.*, vol. 18, no. 6, pp. 930–936, Dec. 2010.

[5] J. Kraus, "Free route airspace (FRA) in Europe," *Perner's Contacts*, vol. 6, no. 4, pp. 129–135, 2011.

[6] R. H. Mogford, J. A. Guttman, S. L. Morrow, and P. Kopardekar, "The complexity construct in air traffic control: A review and synthesis of the literature," Federal Aviation Administration Technical Center, Washington, DC, USA, Tech. Rep. DOT/FAA/CT-TN95/22, 1995.

[7] A. S. Bin Jumad, K. Tominaga, C. X. Yi, V. N. Duong, E. Itoh, and M. Schultz, "Flow-centric air traffic control: Human in the loop simulation experiment," in *Proc. IEEE/AIAA 42nd Digit. Avionics Syst. Conf. (DASC)*, Oct. 2023, pp. 1–8.

[8] H. Zheng, Z. Wang, C. Zheng, Y. Wang, X. Fan, W. Cong, and M. Hu, "A graph multi-attention network for predicting airport delays," *Transp. Res. E, Logistics Transp. Rev.*, vol. 181, Jan. 2024, Art. no. 103375.

[9] K. Sekine, T. Tatsukawa, K. Fujii, and E. Itoh, "Air traffic simulation across fir in Japan using cellular automaton," in *Proc. 33rd Congr. Int. Council Aeronaut. Sci.*, 2024, pp. 6502–6518.

[10] K. Sekine, T. Tatsukawa, E. Itoh, and K. Fujii, "Multi-objective takeoff time optimization using cellular automaton-based simulator," *IEEE Access*, vol. 9, pp. 79461–79476, 2021.

[11] K. Sekine, F. Kato, T. Tatsukawa, K. Fujii, and E. Itoh, "Rule design for interpretable en route arrival management via runway-flow and inter-aircraft control," *IEEE Access*, vol. 11, pp. 75093–75111, 2023.

[12] K. Sekine, T. Tatsukawa, T. Aoki, K. Fujii, and E. Itoh, "Adaptive and interpretable speed control rules for en route arrival management using extended learning classifier system," *IEEE Access*, vol. 13, pp. 39631–39649, 2025.

[13] A. M. Bayen, R. L. Raffard, and C. J. Tomlin, "Adjoint-based control of a new Eulerian network model of air traffic flow," *IEEE Trans. Control Syst. Technol.*, vol. 14, no. 5, pp. 804–818, Sep. 2006.

[14] D. Sun and A. M. Bayen, "Multicommodity Eulerian–Lagrangian large-capacity cell transmission model for en route traffic," *J. Guid., Control, Dyn.*, vol. 31, no. 3, pp. 616–628, May 2008.

[15] P. Wei, Y. Cao, and D. Sun, "Total unimodularity and decomposition method for large-scale air traffic cell transmission model," *Transp. Res. B, Methodol.*, vol. 53, pp. 1–16, Jul. 2013.

[16] K. Sekine, F. Kato, K. Kageyama, and E. Itoh, "Data-driven simulation for evaluating the impact of lower arrival aircraft separation on available airspace and runway capacity at Tokyo international airport," *Aerospace*, vol. 8, no. 6, p. 165, Jun. 2021.

[17] K. D. Bilimoria, B. Sridhar, S. R. Grabbe, G. B. Chatterji, and K. S. Sheth, "FACET: Future ATM concepts evaluation tool," *Air Traffic Control Quart.*, vol. 9, no. 1, pp. 1–20, Jan. 2001.

[18] S. Alam, H. A. Abbass, and M. Barlow, "ATOMS: Air traffic operations and management simulator," *IEEE Trans. Intell. Transp. Syst.*, vol. 9, no. 2, pp. 209–225, Jun. 2008.

[19] Transoft Solutions. (2025). *AirTOP–Fast-Time Simulation for Airport and Airspace*. Accessed: May 28, 2025. [Online]. Available: https://www.transoftsolutions.com/ja/aviation/software/airport-airspace-fast-time-simulation/airtop-overview/

[20] D. Sun, I. S. Strub, and A. M. Bayen, "Comparison of the performance of four Eulerian network flow models for strategic air traffic management," *Netw. Heterogeneous Media*, vol. 2, no. 4, pp. 569–595, 2007.

[21] Y. Guleria, Q. Cai, S. Alam, and L. Li, "A multi-agent approach for reactionary delay prediction of flights," *IEEE Access*, vol. 7, pp. 181565–181579, 2019.

[22] G. Gurtner, L. Delgado, and D. Valput, "An agent-based model for air transportation to capture network effects in assessing delay management mechanisms," *Transp. Res. C, Emerg. Technol.*, vol. 133, Dec. 2021, Art. no. 103358.

[23] X. Zhao, Y. Wang, L. Li, and D. Delahaye, "A queuing network model of a multi-airport system based on point-wise stationary approximation," *Aerospace*, vol. 9, no. 7, p. 390, Jul. 2022.

[24] E. Itoh, M. Mitici, and M. Schultz, "Modeling aircraft departure at a runway using a time-varying fluid queue," *Aerospace*, vol. 9, no. 3, p. 119, Feb. 2022.

[25] K. Higasa and E. Itoh, "Controlling aircraft inter-arrival time to reduce arrival traffic delay via a queue-based integer programming approach," *Aerospace*, vol. 9, no. 11, p. 663, Oct. 2022.

[26] K. Higasa, K. Sekine, and E. Itoh, "Effectiveness of aircraft inter-arrival control in upstream traffic flow via a combined tandem fluid queue model and integer programming approach," *IEEE Access*, vol. 11, pp. 15252–15270, 2023.

[27] E. Itoh, K. Tominaga, M. Schultz, and V. N. Duong, "Untangling complexity in ASEAN air traffic management through time-varying queuing models," *Aerospace*, vol. 11, no. 1, p. 11, Dec. 2023.

[28] R. Barhydt, T. M. Eischeid, M. T. Palmer, and D. J. Wing, "Regaining lost separation in a piloted simulation of autonomous aircraft operations," in *Proc. 5th USA/Europe Air Traffic Manage. R&D Seminar*, 2002, pp. 1–11.

[29] S. B. Amor, H. T. Dac, M. Bui, and V. Duong, "Simulating dynamic atm network effects using cellular automata," in *Proc. Eur. Conf. Complex Syst.*, Dresden, Germany, 2007, Paper 149.

[30] R. Mori, "Aircraft ground-taxiing model for congested airport using cellular automata," *IEEE Trans. Intell. Transp. Syst.*, vol. 14, no. 1, pp. 180–188, Mar. 2013.

[31] F. Mazur and M. Schreckenberg, "Simulation and optimization of ground traffic on airports using cellular automata," *Collective Dyn.*, vol. 3, no. 1, Mar. 2018, pp. 1–22.

[32] S. Tsuzuki, D. Yanagisawa, and K. Nishinari, "Throughput reduction on an air-ground transport system by the simultaneous effect of multiple traveling routes equipped with parking sites," *J. Phys. Commun.*, vol. 4, no. 5, May 2020, Art. no. 055009.

[33] Y. Kawagoe, R. Chino, S. Tsuzuki, E. Itoh, and T. Okabe, "Analyzing stochastic features in airport surface traffic flow using cellular automaton: Tokyo international airport," *IEEE Access*, vol. 10, pp. 95344–95355, 2022.

[34] C. Kim, K. Abubakar, and O. Obah, "Cellular automata modeling of en route and arrival self-spacing for autonomous aircrafts," in *Proc. 50th Annu. Meeting Air Traffic Controllers Assoc.*, 2005, pp. 127–134.

[35] S.-P. Yu, X.-B. Cao, and J. Zhang, "A real-time schedule method for aircraft landing scheduling problem based on cellular automation," *Appl. Soft Comput.*, vol. 11, no. 4, pp. 3485–3493, Jun. 2011.

[36] Y. He, K. Cai, Y. Li, and M. Xiao, "An improved cellular-automaton-based algorithm for real-time aircraft landing scheduling," in *Proc. 7th Int. Symp. Comput. Intell. Design*, vol. 1, Dec. 2014, pp. 284–288.

[37] S.-J. Wang and Y.-H. Gong, "Research on air route network nodes optimization with avoiding the three areas," *Saf. Sci.*, vol. 66, pp. 9–18, Jul. 2014.

[38] W.-X. Lim and Z.-W. Zhong, "Re-planning of flight routes avoiding convective weather and the 'Three areas,'" *IEEE Trans. Intell. Transp. Syst.*, vol. 19, no. 3, pp. 868–877, Mar. 2018.

[39] F. Enayatollahi and M. A. A. Atashgah, "Wind effect analysis on air traffic congestion in terminal area via cellular automata," *Aviation*, vol. 22, no. 3, pp. 102–114, Nov. 2018.

[40] F. Enayatollahi, M. A. A. Atashgah, S. M. Malaek, and P. Thulasiraman, "PBN-based time-optimal terminal air traffic control using cellular automata," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 57, no. 3, pp. 1513–1523, Jun. 2021.
[41] H. Yun-Xiang and H. Xiao-Qiong, "Modeling of air traffic flow using cellular automata," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 58, no. 4, pp. 2623–2631, Aug. 2022.
[42] I. I. Ogedengbe, T. S. Tai, K. Y. M. Wong, and R. P. Liem, "Cellular automata for the investigation of navigation dynamics and aircraft mix in terminal arrival traffic," *Phys. A, Stat. Mech. Appl.*, vol. 671, Aug. 2025, Art. no. 130628.
[43] Y. Jianzhong, H. Haoran, Y. Wang, M. Xiaozhen, H. Minghua, D. Delahaye, and S. Alam, "Robust pre-departure scheduling for a nation-wide air traffic flow management," *Chin. J. Aeronaut.*, vol. 38, no. 4, Apr. 2025, Art. no. 103223.
[44] Electronic Navigation Research Institute. (2025). *CARATS Open Data Related Information*. Accessed: Jun. 11, 2025. [Online]. Available: https://www.enri.go.jp/en/caratsopendata.html
[45] *User Manual for the Base of Aircraft Data (BADA)*, EUROCONTROL, Brussels, Belgium, 2019.
[46] Japan Aeronautical Information Service Center. (2024). *Aeronautical Information Publication (AIP)*. Accessed: Jan. 5, 2024. [Online]. Available: https://aisjapan.mlit.go.jp/
[47] Research Institute for Sustainable Humanosphere, Kyoto University. (2020). *Meso-Scale Model Grid Point Value (MSMGPV)*. Accessed: Sep. 1, 2022. [Online]. Available: http:// database.rish.kyoto-u.ac.jp/index-e.html
[48] R. Dalmau, M. M. Farrés, and X. P. Menéndez, "PyBADA: Easy BADA integration in Python for rapid prototyping," in *Proc. SESAR Innov. Days, Posters Abstr.*, 2018, pp. 16–17.
[49] K. Sekine, D. Iwata, P. Bouchaudon, T. Tatsukawa, K. Fujii, K. Tominaga, and E. Itoh, "Validating flow-based arrival management for en route airspace: Human-in-the-loop simulation experiment with ESCAPE light simulator," *Aerospace*, vol. 11, no. 11, p. 866, Oct. 2024.



**KATSUHIRO SEKINE** received the B.E. and M.E. degrees in industrial management and engineering from The Tokyo University of Science, Japan, in 2019 and 2022, respectively, and the Ph.D. degree in engineering from the Department of Information and Computer Technology, The Tokyo University of Science, in 2025. During his undergraduate career, he visited the German Aerospace Center (DLR) as a Research Trainee and conducted research on extended arrival management. He is currently a Research Trainee with the Air Traffic Management Department, Electronic Navigation Research Institute, National Institute of Maritime, Port and Aviation Technology. Since April 2025, he has been an Assistant Professor with the Research Center for Advanced Science and Technology, The University of Tokyo. His research interests include multiobjective optimization, statistical and machine-learning methods, modeling and simulation, and their applications in air traffic management.



**TOMOAKI TATSUKAWA** (Member, IEEE) received the B.S. and M.S. degrees in engineering from Tokyo Institute of Technology, Tokyo, Japan, in 2002 and 2004, respectively, and the Ph.D. degree in engineering from The University of Tokyo, in 2012. He was a Software Engineer with SGI Japan Ltd., from 2004 to 2010. From 2012 to 2015, he was a Postdoctoral Research Fellow with the Institute of Space and Astronautical Science, Japan Aerospace Exploration Agency (ISAS/JAXA), Sagamihara, Japan. From 2015 to 2019, he was a Junior Associate Professor with Tokyo University of Science, Tokyo. Since April 2020, he has been an Associate Professor with the Department of Information and Computer Technology, Tokyo University of Science. Since April 2024, he has been a Professor with the Department of Information and Computer Technology. His research interests include multiobjective evolutionary computation, aerodynamic design optimization, and multiobjective design exploration. He is a member of JSME. He received the Young Researcher Award from the IEEE Computational Intelligence Society, Japan Chapter, in 2011.



**YOSHIAKI KAWAGOE** received the Ph.D. degree in engineering from Tohoku University, Japan, in 2018. From 2018 to 2020, he was a Specially Appointed Assistant Professor with the Institute of Fluid Science, Tohoku University. From April 2020 to March 2025, he was an Assistant Professor with the Department of Aerospace Engineering, Tohoku University. Since April 2025, he has been a Visiting Associate Professor with the Department of Aerospace Engineering. Since April 2025, he has been a Specially Appointed Associate Professor with the Department of Aeronautics and Astronautics, The University of Tokyo. His research interests include multiscale modeling of composite materials for next-generation aircraft and optimization of surface traffic flow at airports.



**ERI ITOH** received the Ph.D. degree in aeronautics and astronautics from The University of Tokyo, Japan, in 2007. After gaining experience in international research organizations, she currently holds the position of a Professor with the Research Center for Advanced Science and Technology, The University of Tokyo, and the Aeronautics and Astronautics Department, The University of Tokyo, and the Chief Researcher with the Air Traffic Management Department, Electronic Navigation Research Institute, National Institute of Maritime, Port and Aviation Technology. Her research interests include the design of automation systems that work with human operators in air-traffic management, including airspace and airport operations. By combining data-driven analyses, mathematical models, and simulation studies, she worked toward realizing even more efficient and resilient air traffic operations.